

2 UNIT MATHEMATICS FORM VI

Time allowed: 3 hours (plus 5 minutes reading)

Exam date: 8th August 2001

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new 4-leaf answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet.

QUESTION ONE (Start a new answer booklet)

Marks

- [1] (a) Convert $\frac{4\pi}{5}$ to degrees.
- [1] (b) Write down a primitive of $\sec^2 5x$.
- [2] (c) The line $5x - ky = 7$ passes through the point $(1, 1)$. Find the value of k .
- [2] (d) Differentiate $y = 5x^3 - 2x + 9$ with respect to x .
- [2] (e) Express $\frac{4}{\sqrt{3}-1}$ with a rational denominator in simplest form.
- [2] (f) Find the exact value of $\tan \frac{\pi}{3} + \tan \frac{\pi}{4}$.
- [2] (g) Solve $|x - 1| = 11$.

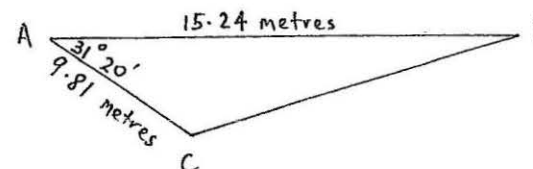
QUESTION TWO (Start a new answer booklet)

Marks

- (a) Differentiate the following with respect to x :
- [2] (i) $x^2 e^x$,
- [2] (ii) $\ln(3x - 2)$,
- [2] (iii) $\sin^2 x$.
- [2] (b) Find a primitive function of $(3x - 4)^6$.
- (c) Evaluate the following definite integrals:
- [2] (i) $\int_1^2 6x^2 dx$,
- [2] (ii) $\int_0^{\frac{\pi}{2}} \sin 2x dx$.

QUESTION THREE (Start a new answer booklet)

(a)

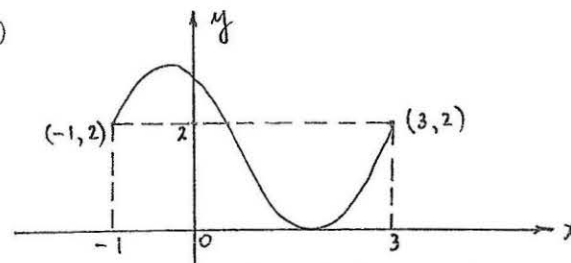


Marks

- [2] (i) Find the length of BC correct to the nearest centimetre.
- [2] (ii) Find the area of $\triangle ABC$ correct to the nearest square metre.
- (b) Consider the geometric series $1 - \frac{1}{3} + \frac{1}{9} - \dots$.
- [1] (i) Explain why the series has a limiting sum.
- [1] (ii) Find the limiting sum.
- [3] (c) Find the equation of the tangent to the curve $y = \ln x$ at the point $(e, 1)$.
- (d) If α and β are roots of the equation $x^2 + 8x + 11 = 0$, find:
- [1] (i) $\alpha + \beta$,
- [1] (ii) $\alpha\beta$,
- [1] (iii) $\alpha^2 + \beta^2$.

QUESTION FOUR (Start a new answer booklet)

(a)

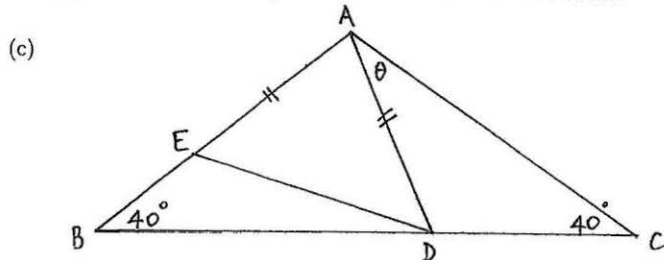


Marks

- [1] (i) Sketch the graph of $y = f(x) + 2$.
- [1] (ii) Given that $\int_{-1}^3 f(x) dx = \frac{15}{2}$, evaluate $\int_{-1}^3 (f(x) + 2) dx$.

(b) A particle moves in a straight line so that its displacement x metres at time t seconds is given by $x = 2t^3 - t^2$.

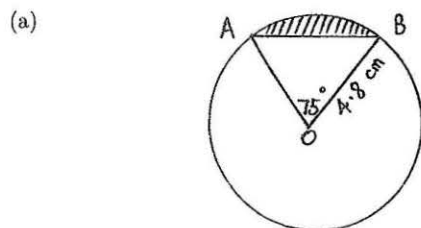
- [2] (i) At what times is the particle at rest?
- [2] (ii) How far does the particle travel between these times?



In the diagram above, $\triangle ABC$ is isosceles with $\angle B = \angle C = 40^\circ$, and $AD = AE$. Let $\angle DAC = \theta$.

- [1] (i) Explain why $\angle ADB = 40^\circ + \theta$.
- [2] (ii) Find an expression for $\angle DAE$ in terms of θ .
- [3] (iii) Show that $\angle EDB = \frac{1}{2}\theta$.

QUESTION FIVE (Start a new answer booklet)

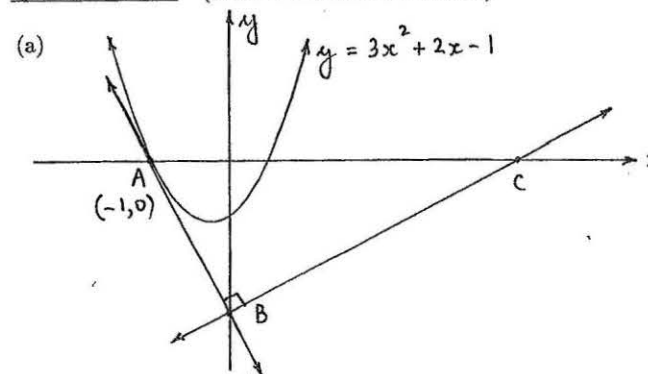


In the diagram above, O is the centre of a circle of radius 4.8 centimetres, and $\angle AOB = 75^\circ$.

- Marks [2] (i) Find the exact length of arc AB .
- [2] (ii) Find the exact area of the sector AOB .
- [2] (iii) Find the area of the minor segment that has been shaded. Give your answer correct to three decimal places.

- [2] (b) (i) Solve $\tan x = -3$ for $0 \leq x \leq 2\pi$. Give your answer in radians correct to three decimal places.
- [2] (ii) On the same diagram, sketch graphs of $y = \tan x$ and $y = -3$ for $0 \leq x \leq 2\pi$.
- [2] (iii) How many solutions are there to the equation $\tan x = -3$ in the domain $-2\pi \leq x \leq 2\pi$?

QUESTION SIX (Start a new answer booklet)



In the diagram above, the graph of $y = 3x^2 + 2x - 1$ and the tangent to the curve at the point $A(-1, 0)$ are drawn.

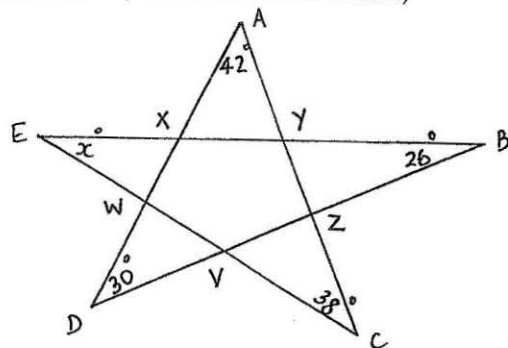
- Marks [2] (i) Show that the equation of the tangent is $y + 4x + 4 = 0$.
- [1] (ii) Show that the tangent meets the y -axis at $B(0, -4)$.
- [2] (iii) Find the equation of the line that passes through B and which is perpendicular to the tangent.
- [1] (iv) Show that this line meets the x -axis at the point $C(16, 0)$.
- [2] (v) Find the area of $\triangle ABC$.
- [4] (b) The region bounded by the curve $y = \tan x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{4}$ is rotated about the x -axis. The volume of the solid formed is given by $V = \pi \int_0^{\frac{\pi}{4}} \tan^2 x \, dx$. Use Simpson's rule with the three function values $x = 0$, $\frac{\pi}{8}$ and $\frac{\pi}{4}$ to approximate the volume. Give your answer correct to three decimal places.

QUESTION SEVEN (Start a new answer booklet)

Marks

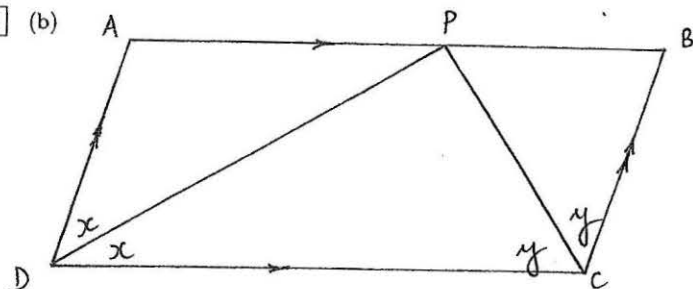
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(a)



Find the value of x . (Give reasons.)

4 (b)



In the diagram above, $ABCD$ is a parallelogram. The point P lies on AB and it is known that $\angle ADP = \angle CDP = x$ and $\angle BCP = \angle DCP = y$.
Prove that $2AD = AB$. (Give reasons.)

1

(c) (i) Write down the discriminant of $5x^2 - 2x + k$.

2

(ii) For what values of k does $5x^2 - 2x + k = 0$ have real roots?

2

(d) Solve $\log_e 16 = 2 \log_e x$.

QUESTION EIGHT (Start a new answer booklet)

Marks

1

(i) How much is in the fund on January 31st 2001?

1

(ii) How much is the first \$1500 deposit worth on December 31st 2010?

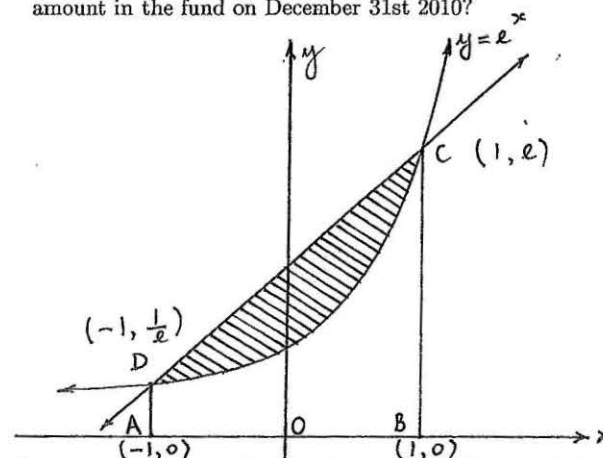
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(iii) Form a geometric series and hence determine the total amount in the fund on December 31st 2010.

2

(iv) If each deposit was increased to \$1600, what difference does it make to the total amount in the fund on December 31st 2010?

(b)



In the diagram above, a straight line has been drawn which intersects with $y = e^x$ at the points $C(1, e)$ and $D(-1, \frac{1}{e})$. The point A has coordinates $(-1, 0)$ and B has coordinates $(1, 0)$. The area between the curves has been shaded.

2

(i) Show that the area of the trapezium $ABCD$ is given by $\frac{e^2 + 1}{e}$.

3

(ii) Hence, or otherwise, find the exact area between the curves.

QUESTION NINE (Start a new answer booklet)

- (a) The value \$ V of a car is given by the formula $V = Ce^{-kt}$, where C and k are constants and t is the time measured in years. Michael bought a car on June 30th 2001 which cost \$65 000 and which was worth \$55 000 after one year.

Marks

- 2** (i) Evaluate the constants C and k .
- 1** (ii) Find the value of the car after 5 years. Give your answer correct to the nearest dollar.
- 2** (iii) In which year will the value of the car fall below half its cost price for the first time?

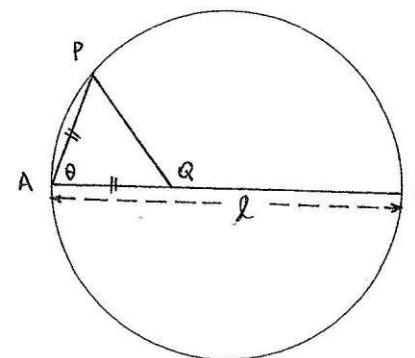
- (b) Concrete is pumped from a truck into a building foundation. The rate $R \text{ m}^3/\text{hour}$ at which the concrete is flowing is given by the expression $R = 9t^2 - t^4$ for $0 \leq t \leq 3$, where t is the time measured in hours after the concrete begins to flow.

- 1** (i) Find the rate of flow at time $t = 2$.
- 1** (ii) Explain why t is restricted to $0 \leq t \leq 3$.
- 3** (iii) Find the maximum flow rate of concrete.
- 2** (iv) When the concrete begins to flow, the foundation has 1000 m^3 already in place. Find an expression for the amount of concrete in the foundation at time t .

QUESTION 10 IS ON THE NEXT PAGE.

QUESTION TEN (Start a new answer booklet)

(a)



In the diagram above, P is a point on the circle with diameter $AB = \ell$. The point Q is on the diameter such that $AP = AQ$. Let $\angle PAQ = \theta$ and let S be the area of $\triangle PAQ$.

Marks

- 3** (i) Show that $S = \frac{\ell^2}{2} \cos^2 \theta \sin \theta$.
- 3** (ii) Find the maximum area of $\triangle APQ$ as P moves along the circumference of the circle.
- 2** (b) (i) Find A and B such that $\frac{1}{(x-1)x} = \frac{A}{x-1} + \frac{B}{x}$.
- 2** (ii) Let $S_n = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(n-1)n}$.
Show that $S_n = 1 - \frac{1}{n}$.
- 2** (iii) Hence or otherwise evaluate $\sum_{n=2}^{\infty} \frac{1}{(n-1)n}$.

JNC

QUESTION 1

a) 144° ✓

b) $\frac{1}{5} \tan 5x + c$ ✓

c) $5 - k = 7$ ✓
 $\therefore k = -2$ ✓

d) $\frac{dy}{dx} = 15x^2 - 2$ ✓✓ (1 each)

e) $\frac{4}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{4(\sqrt{3}+1)}{2}$ ✓
 $= 2(\sqrt{3}+1)$ ✓

f) $\sqrt{3} + 1$ ✓✓ (1 each)

g) $|x-1| = 11$

$x-1 = 11$ or $x-1 = -11$

$\therefore x = 12$ or -10 ✓✓

Grammar 2001
SOLUTIONS

QUESTION 2

a) (i) $\frac{dy}{dx} = 2xe^x + x^2e^x$ ✓✓
 $= xe^x(2+x)$ (not necessary)

(ii) $\frac{dy}{dx} = \frac{3}{3x-2}$ ✓✓

(iii) $\frac{dy}{dx} = 2 \sin x \cdot \cos x$ ✓✓

b) $\frac{(3x-4)^7}{21}$ ✓✓

c) (i) $\int_1^2 6x^2 dx = \left[2x^3 \right]_1^2$ ✓
 $= 2 \times 8 - 2 \times 1$ ✓
 $= 14$

(ii) $\int_0^{\frac{\pi}{2}} \sin 2x dx = -\frac{1}{2} \left[\cos 2x \right]_0^{\frac{\pi}{2}}$ ✓
 $= -\frac{1}{2} [\cos \pi - \cos 0]$
 $= -\frac{1}{2} [-1 - 1]$ ✓
 $= 1$ ✓

QUESTION 3

(a) (i) $BC^2 = 9.81^2 + 15.24^2 - 2 \times 9.81 \times 15.24 \times \cos 31^\circ 20'$ ✓
 $= 73.089... \text{ m}$

$\therefore BC \doteq 8.55 \text{ m}$ (nearest cm) ✓ (Penalise incorrect rounding but ignore no rounding to 2 d.p.)

(ii) Area $\Delta ABC = \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \cdot 9.81 \cdot 15.24 \cdot \sin 31^\circ 20'$ ✓
 $= 38.87... \text{ m}^2$
 $\doteq 39 \text{ m}^2$ ✓

(b) (i) $r = -\frac{1}{3}$. Since $|r| < 1$, S_∞ exists ✓

(ii) $S_\infty = \frac{a}{1-r}$
 $= \frac{1}{\frac{4}{3}}$
 $= \frac{3}{4}$ ✓

(c) $\frac{dy}{dx} = \frac{1}{x}$
 At $x = e$, $\frac{dy}{dx} = \frac{1}{e}$ } ✓ (for either or both)

Equation of tangent is:

$y - 1 = \frac{1}{e}(x - e)$ ✓

$y - 1 = \frac{x}{e} - 1$

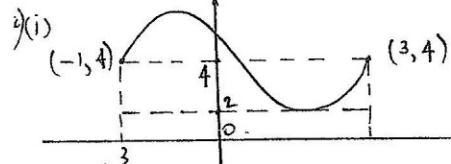
$\therefore x = ey$ ✓

(d) (i) $\alpha + \beta = -8$ ✓

(ii) $\alpha\beta = 11$ ✓

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (-8)^2 - 2 \times 11$
 $= 42$ ✓

QUESTION 4



(ii) $\int_{-1}^3 f(x) + 2 \, dx = \frac{15}{2} + 8$
 $= \frac{31}{2}$ ✓

b) $x = 2t^3 - t^2$ ✓

(i) $\dot{x} = 6t^2 - 2t$ ✓

At rest when $\dot{x} = 0$

$\therefore 2t(3t - 1) = 0$

$\therefore t = 0 \text{ or } \frac{1}{3}$ ✓

(ii) At $t = 0$, $x = 0$.

$t = \frac{1}{3}$, $x = 2 \cdot \frac{1}{27} - \frac{1}{9}$

$= -\frac{1}{27}$ ✓

$\therefore$ Distance travelled is $\frac{1}{27} \text{ m}$. ✓

(c) (i) $\angle ADB = 40^\circ + \theta$ (exterior angle of Δ) ✓

(ii) $\angle DBE = 40^\circ$ (isosceles Δ) ✓

$\angle DAE + 40^\circ + \theta + 40^\circ = 180^\circ$ (angle sum) ✓

$\therefore \angle DAE = 100^\circ - \theta$ ✓

(iii) $\angle ADE = \frac{1}{2}(180^\circ - (100^\circ - \theta))$ (base $\angle$ of isosceles ΔADE) ✓

$= 40^\circ + \frac{\theta}{2}$ ✓

$\therefore \angle EDB = (40^\circ + \theta) - (40^\circ + \frac{\theta}{2})$

$= \frac{\theta}{2}$ ✓

QUESTION 5

a) (i) $75^\circ = 75 \times \frac{\pi}{180}$
 $= \frac{5\pi}{12}$ ✓

$\therefore l = \frac{5\pi}{12} \times 4.8$
 $= 2\pi \text{ cm}$ ✓

(ii) Area of sector $= \frac{1}{2} \cdot (4.8)^2 \cdot \frac{5\pi}{12}$ ✓
 $= \frac{24\pi}{5} \text{ (or } 4.8\pi) \text{ cm}^2$ ✓

(iii) Area of segment $= \text{Area of sector} - \text{Area of } \Delta$
 $= \frac{24\pi}{5} - \frac{1}{2} \cdot (4.8)^2 \cdot \sin \frac{5\pi}{12}$ ✓
 $= 3.952 \text{ cm}^2$ ✓

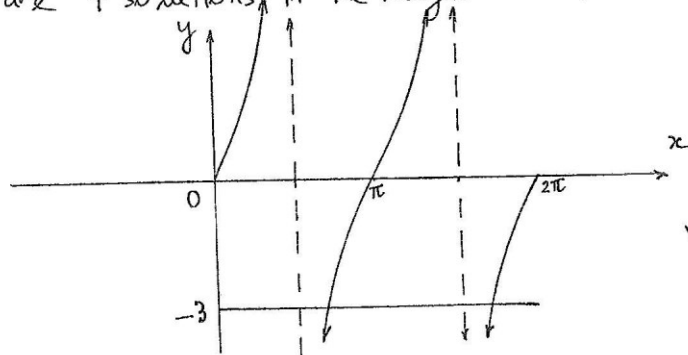
(b) (i) $\tan x = -3$

Related angle $= 1.249$ ✓

$\therefore x = \pi - 1.249 \text{ or } 2\pi - 1.249$
 $= 1.893 \text{ or } 5.034$ ✓

(need both for this mark. No marks for degree equivalent)

(ii) Since there are 2 solutions from $0 \leq x \leq 2\pi$, there are 4 solutions in the range $-2\pi \leq x \leq 2\pi$ ✓



✓✓ (Do not penalise if they draw graphs outside domain)

QUESTION 6

a) (i) $y = 3x + 2x - 1$
 $\frac{dy}{dx} = 6x + 2$

At $x = -1$, $\frac{dy}{dx} = -4$ ✓

Eqn is $y - 0 = -4(x + 1)$ ✓
 $\Rightarrow y + 4x + 4 = 0$ ✓

(ii) When $x = 0$, $y + 4 = 0$

$\therefore y = -4$

So $B = (0, -4)$ ✓

(iii) $M = \frac{1}{4}$ ✓

$y + 4 = \frac{1}{4}(x - 0)$ ✓

$\Rightarrow 4y + 16 = x$ ✓

(iv) When $y = 0$, $x = 16$

$\therefore C = (16, 0)$ ✓

(v) Area $\Delta ABC = \frac{1}{2} \times 4 \times 16$ ✓

$= 32 \text{ units}^2$ ✓

(b)

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$
y	0	0.17157	1

$V = \pi \int_0^{\frac{\pi}{4}} \tan^2 x \, dx$

$= \pi \left[\frac{\pi}{24} (0 + 4 \times 0.17157 + 1) \right]$ ✓✓

$= 0.693 \text{ (u}^3)$ ✓✓

(subtract 1 mark for incorrect rounding or if they have not attempted to round off)

QUESTION 7

- a) $\angle BWA = 80^\circ$ (external $\angle$ of $\triangle AWD$) ✓
 $\angle BXC = 56^\circ$ (external $\angle$ of $\triangle CXE$) ✓
 $x = 180 - (80 + 56)$ ($\angle$ sum of $\triangle$) ✓
 $\therefore x = 44$ ✓
- b) $\angle DPA = x$ (alternate $\angle$ on || lines) ✓
 $\therefore \triangle ADP$ is isosceles $\Rightarrow AD = AP$ — 1. ✓
 $\angle BPC = y$ (alternate $\angle$ on || lines) ✓
 $\therefore \triangle BCP$ is isosceles $\Rightarrow BC = BP$ — 2. ✓
 Since ABCD is a parallelogram, $AB = BC$. — 3.

By 1, 2 & 3; $AD = AP = BC = BP$.

$$\therefore AB = AP + BP$$

$$= AD + AD$$

$$= 2AD, \text{ as required.}$$

✓✓ (There are other acceptable methods. Allocate marks similarly if possible)

- c) (i) $\Delta = 2 - 4.5k$ ✓
 $= 4 - 20k$ ✓
- (ii) For real roots, $\Delta \geq 0$ } ✓ (either)
 $4 - 20k \geq 0$ }
 $\therefore k \leq \frac{1}{5}$ ✓

- d) $\ln 16 = 2 \ln x$ ✓
 $\therefore 2 \ln 4 = 2 \ln x$ ✓
 $\therefore x = 4$ ✓

[or $\ln 16 = \ln x^2$ ✓
 $\therefore x^2 = 16$ ✓
 $x = \pm 4$
 but $x > 0 \therefore x = 4$ only] ✓

QUESTION 8

a) (i) $A_1 = 1500 (1.0075)^{120}$ ✓
 $= \$1511.25$ ✓

(ii) $A = 1500 (1.0075)^{120}$ ✓
 $= \$3677.04$ ✓

(iii) Total Amount = $1500 [1.0075^{120} + 1.0075^{119} + 1.0075^{118} + \dots + 1.0075]$ ✓
 $= 1500 \left[1.0075 \frac{(1.0075^{120} - 1)}{(1.0075 - 1)} \right]$ ✓

$$= 1500 \times 194.9656342$$

$$= \$292448$$
 ✓

(iv) New Total = 1600×194.9656342 } ✓
 $= \$311945$ }

$\therefore$ Difference = $\$19497$ ✓

b) (i) Area ABCD = $\frac{1}{2} \cdot 2 \left(e + \frac{1}{e} \right)$ ✓

$$= \frac{e^2 + 1}{e} \quad (u^2) \quad \checkmark$$

(ii) Area under the curve = $\int_1^e e^x dx$ ✓

$$= [e^x]_1^e$$

$$= e - e^{-1}$$

$$= e - \frac{1}{e}$$

$$= \frac{e^2 - 1}{e} \quad (u^2) \quad \checkmark$$

$\therefore$ Area between the curves = $\frac{e^2 + 1}{e} - \frac{e^2 - 1}{e}$

$$= \frac{2}{e} \cdot (u^2) \quad \checkmark$$

QUESTION 9

a) (i) $V = Ce^{-kt}$
 When $t=0$, $V=65000$
 $\therefore C = 65000$ ✓
 When $t=1$, $V=55000$
 $\therefore 55000 = 65000 e^{-k}$
 $e^{-k} = \frac{11}{13}$

$-k = \ln\left(\frac{11}{13}\right)$
 $\therefore k = -\ln\left(\frac{11}{13}\right)$ ✓
 $\div 0.16705...$

(ii) When $t=5$,
 $V = 65000 e^{-5k}$
 $= \$28194$ ✓

(iii) We need t such that
 $V < \frac{65000}{2}$

$\therefore 65000 e^{-kt} < \frac{65000}{2}$

$e^{-kt} < \frac{1}{2}$
 $-kt < \ln \frac{1}{2}$
 $t > 4.149...$ ✓

$\therefore$ Car falls below half its cost price in 2005 ✓

b) (i) $R = 9t^2 - t^4$
 When $t=2$; $R = 9 \cdot 4 - 2^4$
 $= 20 \text{ m}^3/\text{h}$ ✓

(ii) When $t=3$, $R=0$ and when $t > 3$, $R < 0$ which suggests that concrete is going back into the truck! Since $t \geq 0$ we have $0 \leq t \leq 3$

(iii) $\frac{dR}{dt} = 18t - 4t^3$ and $\frac{d^2R}{dt^2} = 18 - 12t$
 $= 2t(9 - 2t^2)$ ✓

$\frac{dR}{dt} = 0$ when $t=0$ or $\frac{3}{\sqrt{2}}$ or $-\frac{3}{\sqrt{2}}$

Since $t \geq 0$, ignore $t = -\frac{3}{\sqrt{2}}$. When $t = \frac{3}{\sqrt{2}}$, $\frac{d^2R}{dt^2} < 0$.

$\therefore$ So maximum flow rate occurs when $t = \frac{3}{\sqrt{2}}$. $\therefore R = 9 \cdot \frac{9}{2} - \frac{81}{4}$ ✓

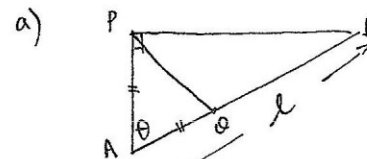
(iv) $A = \int 9t^2 - t^4 dt$
 $A = 3t^3 - \frac{t^5}{5} + C$ ✓

When $t=0$, $A=1000 \therefore C=1000$.

$\therefore A = 3t^3 - \frac{t^5}{5} + 1000$ ✓

(Be generous in part (ii).)

QUESTION 10



(i) $\cos \theta = \frac{AP}{l}$
 $\therefore AP = l \cos \theta$ ✓
 $S = \frac{1}{2} \cdot l \cos \theta \cdot l \cos \theta \cdot \sin \theta$ ✓
 $S = \frac{l^2}{2} \cos^2 \theta \sin \theta$, as required. ✓
 $= \frac{l^2}{2} (1 - \sin^2 \theta) (\sin \theta)$

$= \frac{l^2}{2} (\sin \theta - \sin^3 \theta)$

(ii) $\frac{dS}{d\theta} = \frac{l^2}{2} (\cos \theta - 3\sin^2 \theta \cos \theta) = \frac{l^2}{2} \cos \theta (1 - 3\sin^2 \theta)$ ✓

$\frac{dS}{d\theta} = 0$ when $\cos \theta = 0$ or $\sin \theta = \pm \frac{1}{\sqrt{3}}$

possible
 Since $0 < \theta < 90^\circ$, $\sin \theta = \frac{1}{\sqrt{3}}$ is only solution.

$\sin \theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{3}}$	$\frac{3}{4}$
$\frac{dS}{d\theta}$	+	0	-

$\therefore$ When $\sin \theta = \frac{1}{\sqrt{3}}$, S is a maximum. ✓

$S = \frac{l^2}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{3}} \right)$
 $= \frac{l^2}{2} \left(\frac{3-1}{3\sqrt{3}} \right)$
 $= \frac{l^2}{3\sqrt{3}}$
 $= \frac{\sqrt{3}l^2}{9}$ is the maximum area. ✓

$$b) (i) \frac{1}{(x-1)x} = \frac{Ax + B(x-1)}{(x-1)x}$$

$$\therefore 1 = Ax + Bx - B$$

$$B = -1 \text{ and } A + B = 0$$

$$\therefore A = 1$$

✓✓

$$\therefore \frac{1}{(x-1)x} = \frac{1}{x-1} - \frac{1}{x}$$

$$(ii) S_n = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(n-1) \times n}$$

$$= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n}\right) \quad \checkmark$$

$$= \frac{1}{1} + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \dots + \left(\frac{-1}{n-1} + \frac{1}{n-1}\right) - \frac{1}{n} \quad \checkmark$$

$$= 1 - \frac{1}{n}, \text{ as required.}$$

$$(iii) \sum_{n=2}^{\infty} \frac{1}{(n-1)n} = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots$$

$$= \lim_{n \rightarrow \infty} S_n \quad (\text{from (ii)})$$

$$= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) \quad \checkmark$$

$$= 1. \quad \checkmark$$